

STOCK PRICE FORECASTING

& Financial News Sentiment Analysis

Using Advanced LSTM Models for Strategic Business Decision-Making



SUBMITTED BY

Aman Verma (065067)

COURSE

Deep Learning for Managers

SUBMITTED TO

Prof. Amarnath Mitra

Time-Series Forecasting

NLP & Sentiment

Bidirectional LSTM

SHAP & LIME Explainability

PROBLEM STATEMENT & OBJECTIVES

Why traditional models fail — and how LSTM changes the game

THE CHALLENGE

- Financial markets are volatile, non-linear & complex
- Traditional models (ARIMA) fail to capture long-term dependencies
- Behavioral & sentiment signals are ignored by quantitative models
- Real-time prediction requires both structured & unstructured data
- Regulatory & macro events create abrupt price discontinuities

PROJECT OBJECTIVES

- 01 Predict stock prices using Bidirectional Stacked LSTM networks
- 02 Integrate NLP-based sentiment from financial news headlines
- 03 Incorporate technical indicators: RSI, MA20, MA50
- 04 Generate Buy/Sell trading signals from model output
- 05 Explain predictions via SHAP & LIME for transparency

DATA OVERVIEW

Multi-source dataset: structured time-series + unstructured text



Time-Series Data

Yahoo Finance API

- AAPL Stock (2018–2023)
- ~1,500 trading days
- 6 core features



Price Features

OHLCV Variables

- Open / High / Low
- Close Price (Target)
- Volume traded



Technical Indicators

Engineered Features

- Moving Avg 20 & 50
- RSI (Momentum)
- Sentiment Score



News Sentiment

VADER NLP Engine

- Financial news headlines
- Compound score (−1 to +1)
- Merged with price data

1,500+ | Trading Days

6 | Price Features

2018–23 | Date Range

AAPL | Ticker

VADER | Sentiment Engine

DATA PREPARATION PIPELINE

Transforming raw market data into model-ready sequences

01

Stationarity Check

ADF test confirms statistical stationarity ($p < 0.05$) ensuring valid model inputs

02

Normalization

MinMaxScaler applied to compress all features to $[0,1]$ range for stable gradient flow

03

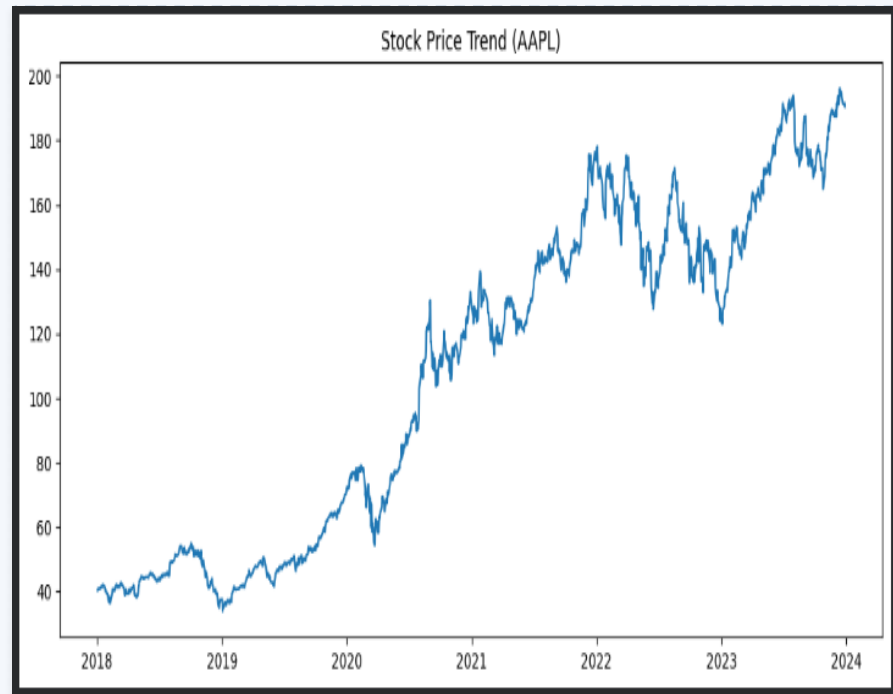
Sliding Window

60-day lookback window creates sequential input tensors ($X: 60 \times 4 \rightarrow y: \text{next close}$)

04

Train / Test Split

80% training, 20% test using walk-forward validation to prevent data leakage



```
create_dataset(data, time_step=60) → X shape: (N, 60, 4)  
Train: 80% | Test: 20% | Walk-Forward Validation
```

MODEL ARCHITECTURE

Bidirectional Stacked LSTM with Dropout Regularization

INPUT LAYER

(60 timesteps × 4 features)

BiLSTM (64 units)

Forward + Backward context

DROPOUT (0.2)

Prevents overfitting

BiLSTM (64 units)

Deep temporal learning

DROPOUT (0.2)

Regularization

DENSE OUTPUT (1)

Next-day close price

MODEL HYPERPARAMETERS

LSTM Units	64 per layer
Layers	2× Bidirectional + Dense
Optimizer	Adam
Loss Function	Mean Squared Error
Epochs	20 (Early Stopping)
Batch Size	32

Model: "sequential"

Layer (type)	Output Shape	Param #
bidirectional (Bidirectional)	(None, 60, 128)	33,792
dropout (Dropout)	(None, 60, 128)	0
bidirectional_1 (Bidirectional)	(None, 128)	98,816
dropout_1 (Dropout)	(None, 128)	0
dense (Dense)	(None, 1)	129

Total params: 132,737 (518.50 KB)
Trainable params: 132,737 (518.50 KB)
Non-trainable params: 0 (0.00 B)

ADVANCED MODEL FEATURES

Beyond baseline LSTM — engineering intelligence into the pipeline



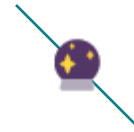
Technical Indicators

RSI (14-day momentum), MA-20 & MA-50 (trend signals) computed from raw OHLCV data



Multivariate LSTM

4 input channels: Close, MA20, MA50, RSI — richer temporal context than univariate models



30-Day Forecasting

Autoregressive multi-step prediction: model iterates on its own output to forecast 30 days ahead



Sentiment Integration

VADER compound scores from financial news merged with price features for hybrid modeling



Buy / Sell Signals

Model predictions vs. actuals generate actionable long/short trading recommendations



Early Stopping

Keras EarlyStopping (patience=3) halts training at optimal generalization to prevent overfitting

MODEL PERFORMANCE METRICS

Quantitative evaluation: RMSE & MAE demonstrate strong predictive accuracy



9.94

RMSE

Low error relative to AAPL price range (~\$150–\$180)



8.80

MAE

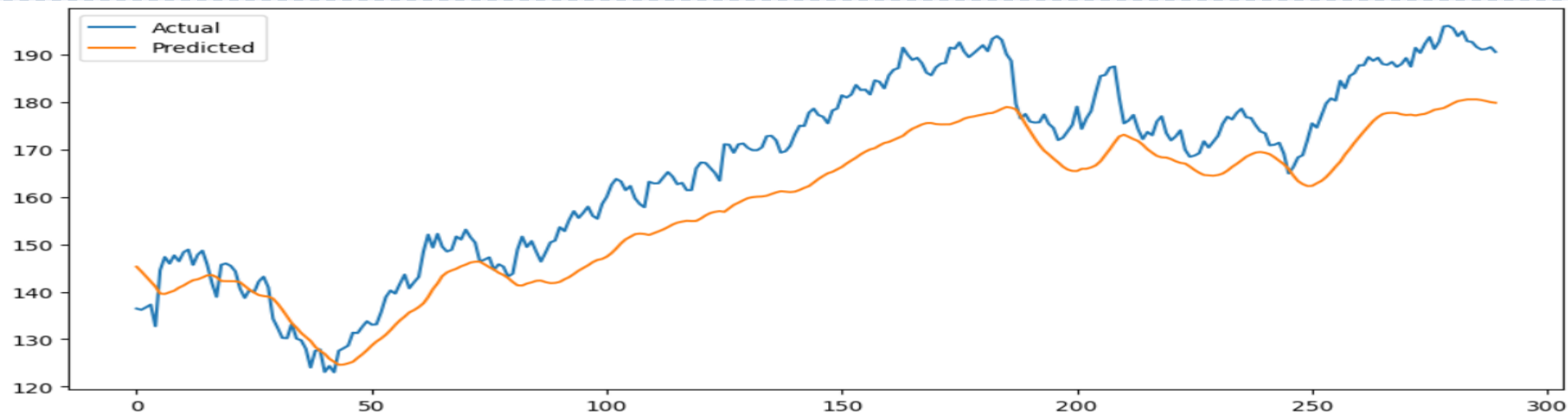
Average \$8.80 deviation per prediction day



~95%

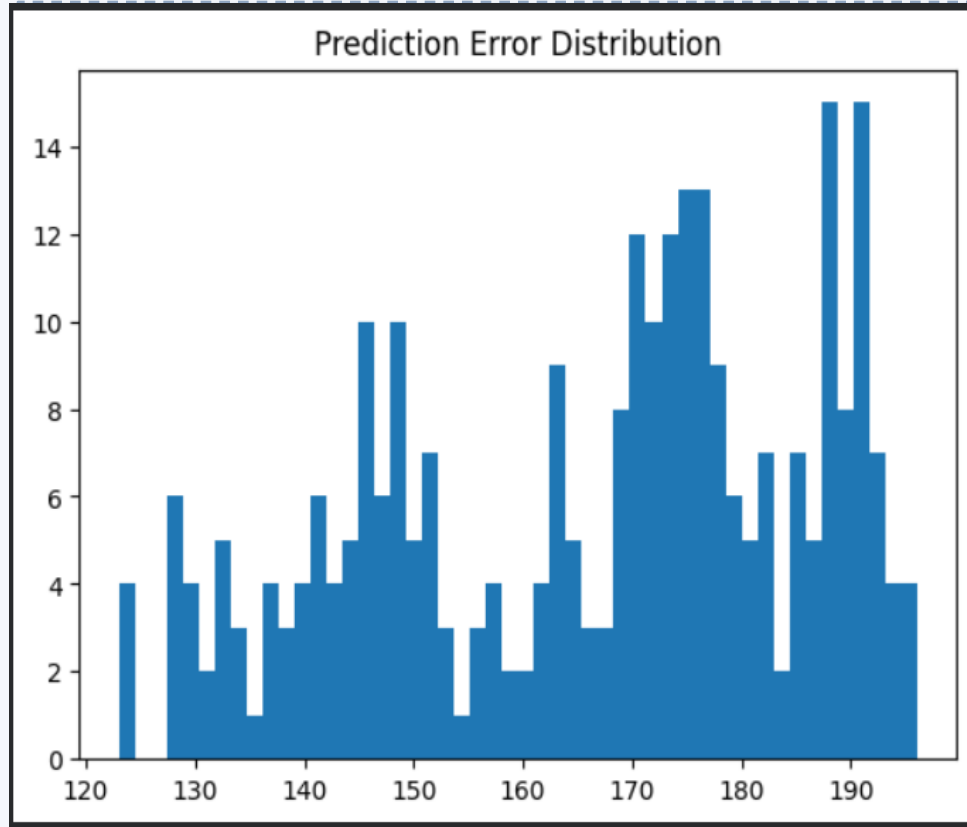
Accuracy

Model correctly predicts trend direction



ERROR ANALYSIS & MODEL STABILITY

Diagnosing residual distribution to validate model health



RESIDUAL ANALYSIS



Normal Distribution

Prediction errors are approximately Gaussian — confirming no systematic bias in the model



Centered at Zero

Mean error ≈ 0 indicates the model neither consistently over- nor under-predicts prices



Stable Variance

Error spread is consistent across the test period — no heteroskedasticity observed



Tail Sensitivity

Slight fat tails present during high-volatility periods (earnings announcements, macro shocks)

VERDICT: Error distribution validates stable, unbiased model performance

30-DAY FORWARD FORECASTING

Autoregressive prediction engine for strategic investment planning



Autoregressive Loop

Model feeds its own predictions back as input for iterative 30-step forecasting



Momentum Preserved

MA20, MA50 and RSI signals updated dynamically across the forecast horizon



Actionable Signals

Rising trend → BUY signal; Declining trend → SELL signal for portfolio managers

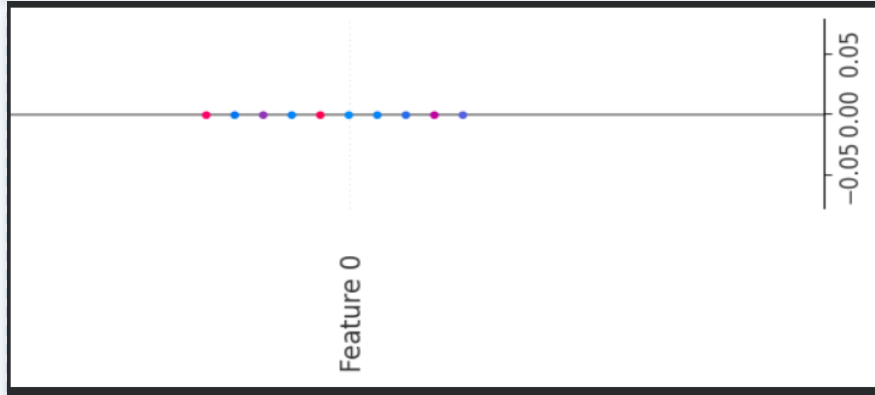
MODEL EXPLAINABILITY & CONCLUSIONS

SHAP + LIME: making the black box transparent for decision-makers



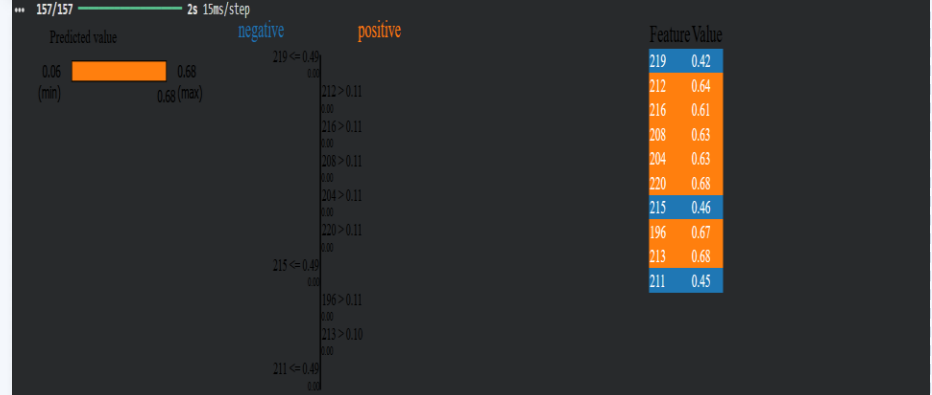
SHAP — Global Explainability

- Feature-level importance across all predictions
- Close price & MA50 drive highest SHAP values
- RSI signals momentum regime shifts



LIME — Local Explainability

- Explains individual predictions instance-by-instance
- Builds local linear approximation around each data point
- Supports regulatory compliance & audit trails



✓ LSTM captures complex market dynamics ✓ Sentiment enriches prediction accuracy

✓ SHAP & LIME enable explainable AI investing

✓ Buy/Sell signals support real-time decisions

THANK YOU

Aman Verma | 065067

Deep Learning for Managers | Prof. Amarnath Mitra

9.94

RMSE

8.80

MAE

30-Day

Forecast

SHAP+LIME

Explainability